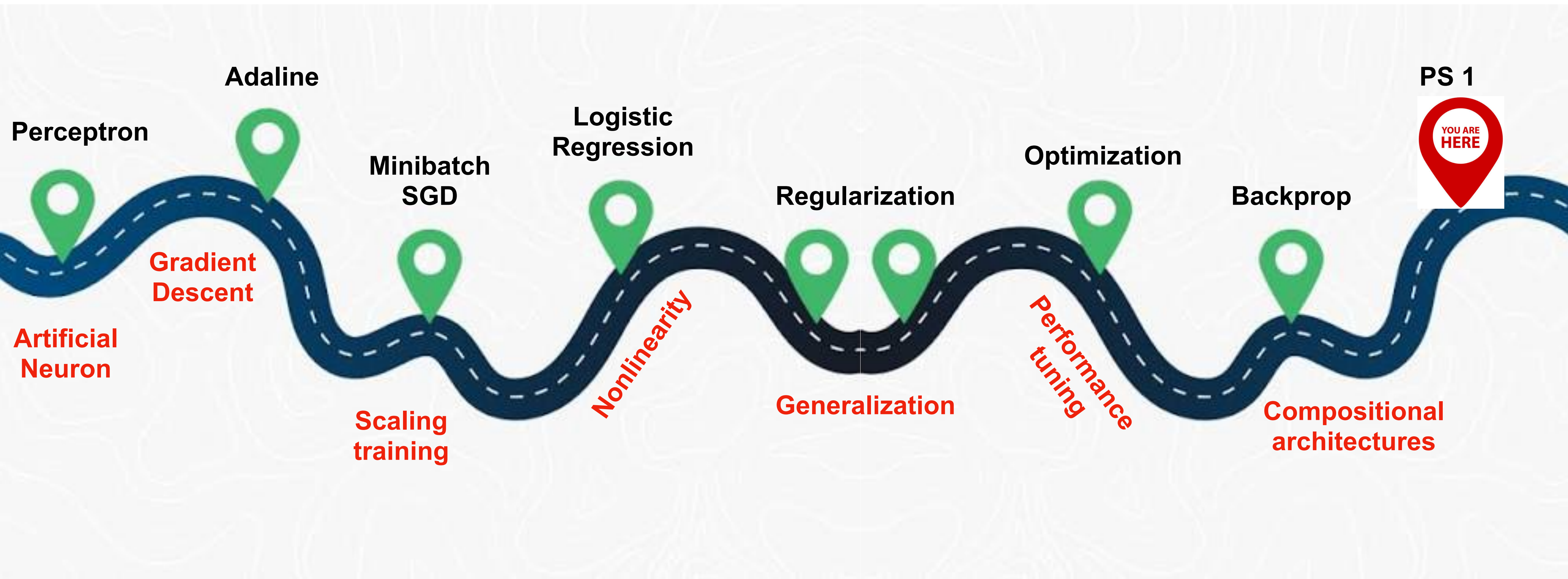


Hands-on with PyTorch

DATA 442 / 610 Lecture 7



Roadmap to Deep Learning



Outline

- Review of backprop, N-layer MLP
- Hands-on with PyTorch

Relevant Reading: *Deep Learning*, Chapter 6

Gradient Descent with a MLP

Goal: Find $\mathbf{w}$ to minimize the loss $\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y^i - y_{NN}^i)^2$

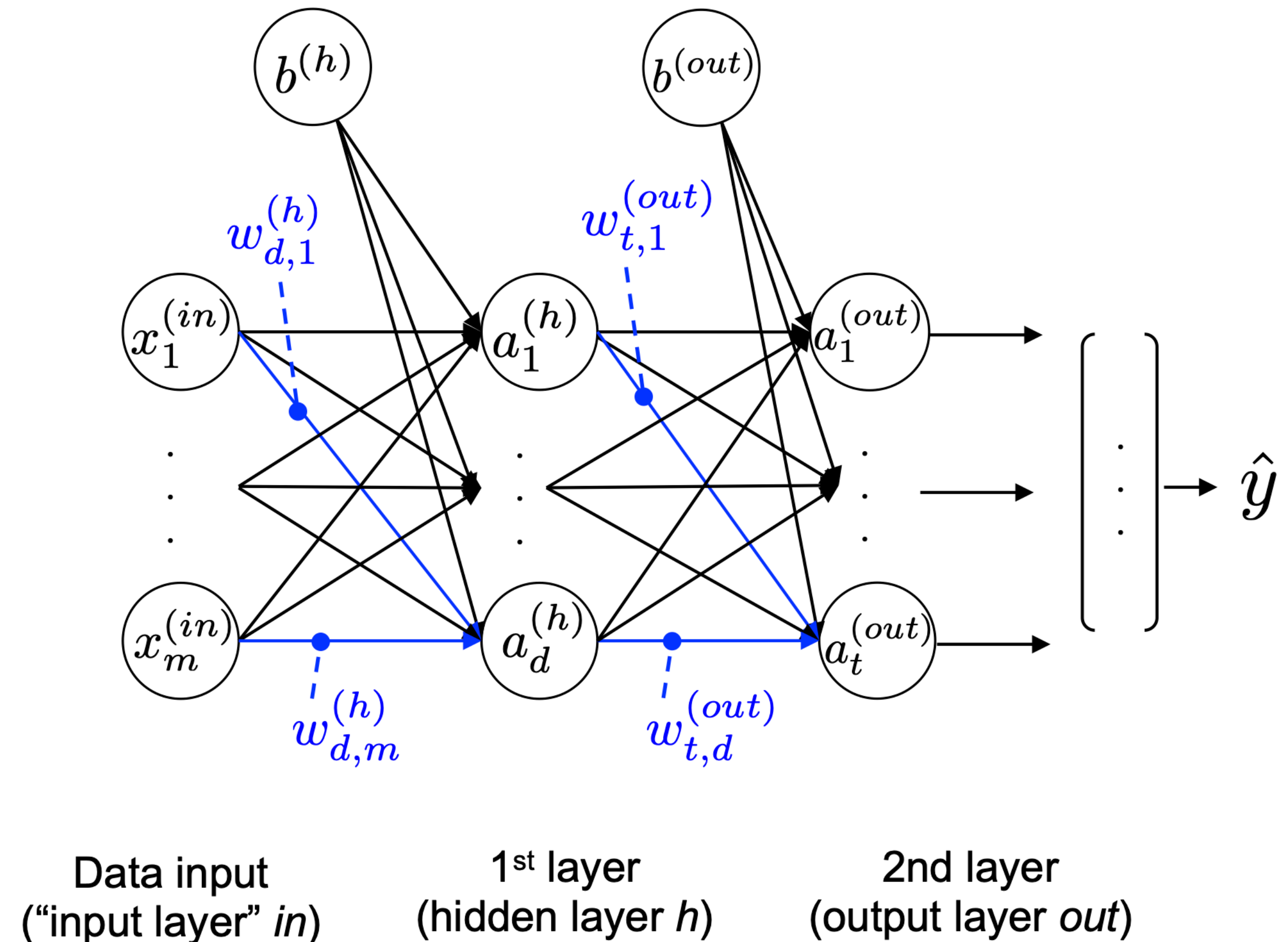
Algorithm: For each training example $\mathbf{x}^i$

1. Compute the output $y_{NN}^i = \sigma(\mathbf{w}_o^\top \sigma(\mathbf{w}_h^\top \mathbf{x}^i + \mathbf{b}_h) + \mathbf{b}_o)$
2. Update the weights and bias

$$w_j \leftarrow w_j + \Delta w_j$$

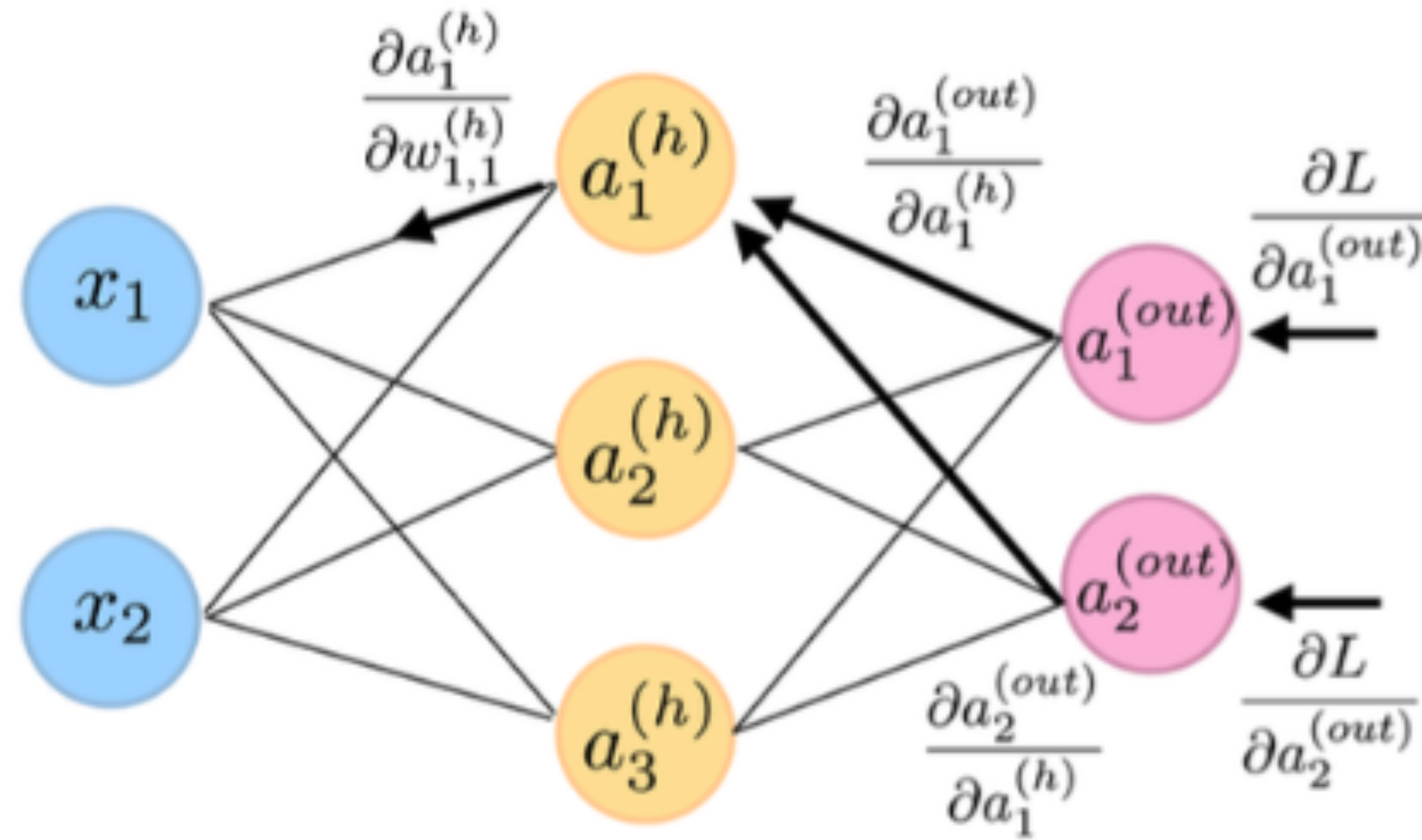
$$b \leftarrow b + \Delta b$$

$$\text{Chain rule: } \frac{d}{dx} f(g(x)) = \frac{df}{dg} \frac{dg}{dx}$$



Courtesy of C. Fanelli

Backpropagation



Backpropagation is an efficient way to calculate gradients by proceeding layer-by-layer

Hands-on Demo: PyTorch



That's all!

- Slides will be posted on the course website: <https://colen-teaching.github.io/DATA442-Fall-2026>
- Lab demos will also be posted
- **PS1** due October 2